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GALAXY CLUSTERS

1 Virgo cluster

The closest major galaxy cluster to Earth, the Virgo Cluster contains perhaps 2,000 galaxies scattered over the constellations Virgo and Coma Berenices. Its larger galaxies are a mix of spirals and ellipticals (with the latter formed by collisions between the former). Its largest member is the giant elliptical galaxy M87, which lies at the center of the cluster, about 53 million light-years from Earth.

2 Abell 383

The dense cluster Abell 383 lies about 2.5 billion light-years away and is a powerful source of X-rays, thanks to a vast cloud of superheated gas stripped away from its individual galaxies. The cluster's enormous mass allows it to act as a gravitational lens, bending the space around it and deflecting the path of light rays from more distant galaxies to produce distorted arcs of light.

3 Stephan's Quintet

This group of five galaxies in Pegasus is deceptive. While four of its galaxies form a tight physical group some 290 million light-years from Earth, the blue spiral at upper left is actually a much nearer foreground object. The other four members—three spirals and an elliptical—will almost certainly merge into a single giant elliptical galaxy in the next billion years or so.

4 M00 J1142+1527

At a distance of 8.5 billion light-years from Earth, this monster cluster is so distant that its light has been redshifted (see p.72) almost to invisibility. As a result it was only discovered in 2015 when observations from two separate infrared space telescopes were combined. With a similar mass to El Gordo (see right), it may be one of just a handful of giant clusters that formed in the first few billion years of cosmic history.